

Security Administration

Objectives

After completing this lesson, you should be able to:

- Describe the `chroot` jail
- Use the `chroot` utility
- Describe packet-filtering firewalls
- Describe `firewalld`
- Configure `firewalld` packet filters
- Describe `iptables`
- Configure `iptables` packet filters
- Describe TCP wrappers
- Configure TCP wrappers
- Describe security control in Oracle Cloud Infrastructure



chroot Jail: Overview

- The `chroot` utility changes the apparent root directory.
 - A program (process) runs with a root directory other than `/`.
 - The artificial root directory is called a `chroot` jail.
- To the process, it appears that it is running in the root directory.
- A `chroot` jail limits the directory access of a potential attacker.
- A `chroot` jail is not intended to:
 - Defend against intentional tampering by privileged (`root`) users
 - Block low-level access to system devices by privileged users
- The `chroot` jail directory must be populated with all files required by the process at their expected locations.

chroot Utility

- To use a `chroot` jail, use the following command:

```
# chroot [OPTION] NEWROOT [COMMAND [ARG]...]
```

- The `NEWROOT` directory becomes the artificial root.
- `chroot` changes to `NEWROOT` and runs the optional command.
 - Alternatively, it runs the `SHELL` variable if the command is omitted.
- Assuming the `NEWROOT` directory exists, the following command fails unless the necessary files are copied into the `NEWROOT` directory before running `chroot`:

```
# chroot /home/oracle/jail  
chroot: failed to run command '/bin/bash': No such file or directory
```

Implementing a `chroot` Jail

- Make the necessary directories and copy all required files into these directories:

```
$ mkdir /home/oracle/jail/bin  
$ cp /bin/bash /home/oracle/jail/bin
```

- Determine whether any shared libraries are required:

```
$ ldd /bin/bash
```

- Create the `lib` (or `lib64`) directory and copy all required shared libraries into this directory:

```
$ mkdir /home/oracle/jail/lib64  
$ cp /lib64/{...} /home/oracle/jail/lib64
```

- Execute the `chroot` command (as `root`):

```
# chroot /home/oracle/jail
```

Running Services in a `chroot` Jail

- DNS and FTP include `chroot` jail options.
- DNS:
 - Install the `bind-chroot` package.
 - `/var/named/chroot` becomes the `chroot` for BIND files.
- FTP (`vsftpd` daemon):
 - Anonymous users are automatically placed in a `chroot` jail.
 - `/var/ftp` appears as `/`.
 - Local user home directories can be configured as `chroot` jails.
 - Set options in the `/etc/vsftpd/vsftpd.conf` file.

Packet-Filtering Firewalls: Introduction

- Packet filtering firewalls accept or deny network packets.
- The Linux kernel has built-in packet filtering functionality.
 - Netfilter is the kernel component that stores filtering rules.
- Two services are available in Oracle Linux 9 to create, maintain, and display the rules stored by Netfilter:
 - `firewalld`
 - `iptables`
- The default firewall service in Oracle Linux 9 is `firewalld`.
- `firewalld` offers several advantages over `iptables`:
 - Changes do not require restart of `firewalld` service.
 - Networks can be separated into different zones based on the level of trust.



firewalld: Introduction

- Firewalld:
 - Is a dynamic firewall manager for Oracle Linux 9
 - Supports firewall (network) zones
 - Supports IPv4, IPv6, and Ethernet bridges
 - Provides a D-BUS interface
 - Provides two configuration modes:
 - **Runtime:** Configuration changes are immediate.
 - **Permanent:** Changes are written to configuration files and are applied when the `firewalld` service restarts.
- Configuration files exist in two directories:
 - `/usr/lib/firewalld`: Contains default configuration files. Do not make changes to these files.
 - `/etc/firewalld`: Configuration changes are written to files in this directory.



firewalld Zones

A `firewalld` zone defines the following firewall features:

- **Services:** Predefined or custom services that are trusted
- **Ports:** Additional ports and associated protocols to trust
- **Protocols:** Allow access for specified protocols
- **Source Ports:** Allow access for specific source ports/port ranges by protocol
- **Masquerading:** Translates IPv4 addresses to a single external address
- **Port Forwarding:** Forward inbound network traffic to an alternative port or IPv4 address
- **ICMP Filter:** Blocks selected ICMP messages
- **Rich Rules:** Extend existing `firewalld` rules to include additional source and destination addresses and logging and auditing actions.
- **Interfaces:** Network interfaces bound to a zone. The zone for an interface is specified with `ZONE=<zone>` in the `/etc/sysconfig/network-scripts/ifcfg` file. If the option is missing, the interface is bound to the default zone.
- **Sources:** Provide source addresses or areas bound to a zone

Predefined firewalld Zones

- The `firewalld` software package includes a set of predefined network zones in the following directory:

```
# ls /usr/lib/firewalld/zones/  
block.xml  drop.xml  home.xml  public.xml  work.xml  
dmz.xml    external.xml  internal.xml  trusted.xml
```

- The zone files contain preset settings, which can be applied to a network interface. For example:

```
# grep -i service /usr/lib/firewalld/zones/public.xml  
<service name="ssh"/>  
<service name="dhcpv6-client"/>
```

- In this example, network interfaces bound to the `public` zone trust only two services, `ssh` and `dhcpv6-client`.

Setting the Default `firewalld` Zone

- After an initial installation, the `public` zone is the default zone as specified in the configuration file, `/etc/firewalld/firewalld.conf`.

```
# grep -i defaultzone /etc/firewalld/firewalld.conf
DefaultZone=public
```

- Network interfaces are bound to the default zone unless specified with `ZONE=<zone>` in the `ifcfg` file.
- You can use the `firewall-cmd` command to change the default zone:

```
# firewall-cmd --set-default-zone=work
success
```

- You can also use the `firewall-config` GUI to change the default zone.

firewalld Services

- A `firewalld` service is a combination of local ports, protocols, modules, and destination addresses.
- A `firewalld` service can also include Netfilter kernel modules that are automatically loaded when a service is enabled.
- The `firewalld` software package includes a set of predefined services in the following directory:

```
# ls /usr/lib/firewalld/services/  
amanda-client.xml ipp-client.xml mysql.xml rpc-bind.xml  
bacula-client.xml ipp.xml      nfs.xml   samba-client.xml  
...
```

- Services can be enabled for a zone in runtime mode.
- Service definitions can be edited only in permanent mode.

Starting firewalld

- To start firewalld:

```
# systemctl start firewalld
```

- To ensure firewalld starts at boot time:

```
# systemctl enable firewalld
```

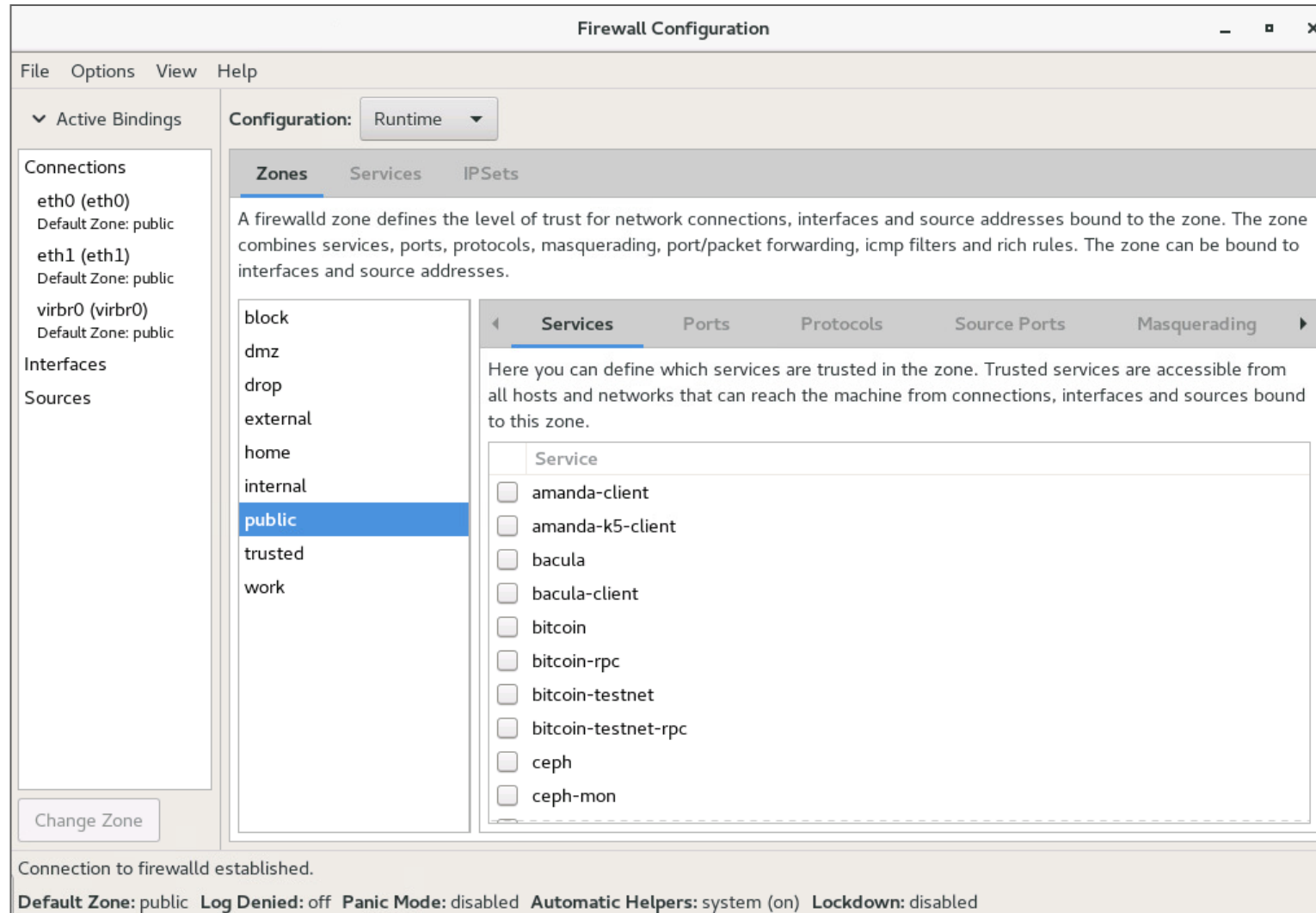
- To check if firewalld is running:

```
# systemctl status firewalld  
# firewall-cmd --state
```

- Three methods to configure the firewalld service:
 - firewall-cmd: Command-line interface
 - firewall-config: Graphical user interface
 - Edit various XML configuration files.

firewalld Configuration Tool

```
# firewall-config
```



firewall-cmd Utility

- The command-line interface to configure the `firewalld` service
- To get help on the `firewall-cmd` command:

```
# firewall-cmd --help
```

- To list information for all zones:

```
# firewall-cmd --list-all-zones
```

- To permit access by HTTP clients for the `public` zone:

```
# firewall-cmd --zone=public --add-service=http
```

- To permanently permit access by HTTP clients for the `public` zone:

```
# firewall-cmd --permanent --zone=public --add-service=http
```

iptables: Introduction

- Another firewall mechanism available in Oracle Linux 9
- Firewall configuration information is stored in the `/etc/sysconfig/` directory:
 - `iptables` file is used for IPv4 configuration.
 - `ip6tables` file is used for IPv6 configuration.
- Every configuration change flushes old firewall rules and reloads new firewall rules.
- Stop the `firewalld` service before using `iptables`.
- Install the `iptables-services` package before attempting to start `iptables` services.
 - `iptables` and `ip6tables` are found here.
- Use the `iptables` command-line utility to create firewall configuration rules.



iptables Terminology

- The Netfilter component is a set of tables:
 - `Filter`: Default table
 - NAT: Network Address Translation table
 - `Mangle`: Table used to alter certain fields in a packet
- Tables store rules, which consist of:
 - One or more match criteria
 - A single action, or target, such as `ACCEPT`, `DROP`, `REJECT`
- Rules are stored in chains. `Filter` table chains are:
 - `INPUT`: Inbound packets pass through this chain.
 - `OUTPUT`: Outbound packets pass through this chain.
 - `FORWARD`: Packets not addressed to the local system pass through this chain.

Beginning iptables Maintenance

- To start the service:

```
# systemctl start iptables
```

- To configure iptables to start at boot time:

```
# systemctl enable iptables
```

- To list iptables:

```
# iptables -L [chain]
```

- Each chain has a default policy:
 - The action to take (ACCEPT or DROP) if no rules match

- To set default policy:

```
# iptables -P chain DROP|ACCEPT
```

- To save configuration changes:

```
# service iptables save
```

Adding a Rule by Using the `iptables` Utility

- To add a rule to a chain, use the following syntax:

```
iptables [-t <table>] -A <chain> <rule_specs> -j <target>
```

- Command-line options and arguments:
 - `-t <table>`: Defaults to the `Filter` table if omitted
 - `-A <chain>`: Appends a rule to `<chain>`
 - `rule_specs`: Specifies the rule criteria
 - `-j <target>`: Specifies the action to take if a match occurs
- Example:

```
# iptables -A INPUT -p tcp -m tcp --dport 80 -j ACCEPT
```

- Accept incoming packets if protocol is TCP and destination port is 80 (`http`).

iptables Rule Specs

- `-p, --protocol`: Matches if the packet uses *protocol*
- `-s, --source address[/mask]`: Matches if the packet came from *address*
- `-d, --destination address[/mask]`: Matches if the packet is going to *address*
- `-j, --jump target`: Specifies what to do if the packet matches the rule specification
- `-i, --in-interface name`: Matches if the packet came from interface *name*
- `-o, --out-interface name`: Matches if the packet is to be sent to interface *name*

NAT Table

- The Netfilter kernel subsystem provides a `nat` table in addition to the default `filter` table to facilitate NAT.
- Use the following option to specify the `nat` table:

```
# iptables -t nat ...
```

- Built-in chains for the `nat` table:
 - PREROUTING: Alters packets when they arrive
 - OUTPUT: Alters locally generated packets before they are sent out
 - POSTROUTING: Alters packets before they are sent out
- Targets for the `nat` table:
 - DNAT: Alters the destination IP address
 - SNAT: Alters the source IP address
 - MASQUERADE: Facilitates use with DHCP

TCP Wrappers

- A TCP wrapper provides basic traffic filtering of incoming network traffic.
- Specifically, a TCP wrapper provides or denies access to “wrapped” network services.
- Use the `ldd` command to determine whether a network service is wrapped (linked to `libwrap.a`):

```
# ldd /usr/sbin/sshd | grep libwrap  
libwrap.so.0 => /lib64/libwrap.so.0 ...
```

- TCP wrappers rely on two configuration files as the basis for access control:
 - `/etc/hosts.allow`
 - `/etc/hosts.deny`
- These files determine whether client access to network service is allowed or denied.

TCP Wrappers Configuration

- Configuration files:
 - `/etc/hosts.allow`: Defines rules that allow client access to server daemons
 - `/etc/hosts.deny`: Defines rules that deny client access to server daemons
- The format for entries is the same for both files:

```
daemon_list : client_list [: command]  
vsftpd : 192.168.2.*
```

- The `/etc/hosts.allow` file is read first:
 - If the daemon-client pair matches, access is granted.
 - The entry in `/etc/hosts.deny` is ignored if the entry in `/etc/hosts.allow` grants access.

More Example

/etc/hosts.allow

```
vsftpd : 192.168.2.*
```

```
sshd : ALL
```

- If daemon is `vsftpd`, allow clients on the `192.168.2` subnet to access FTP
- IF daemon is `sshd`, allow all clients to access `ssh`, `scp`, and `sftp`

/etc/hosts.deny

```
vsftpd : ALL
```

```
vsftpd : .example.com
```

- Deny FTP service to all clients except subnet `192.168.2.*`
- Use the `.domain` syntax to represent any hosts from a given domain. The following example denies connections to `vsftpd` from any host in the `example.com` domain

TCP Wrapper Command Options

- Use the optional *command* argument to send connection banners, warn of attacks, and enhance logging.

- To display the contents of a banner file:

```
vsftpd : ALL : banners /etc/banners/
```

- To append an entry to a log file:

```
ALL : 200.182.68.0 : spawn /bin/echo `date` %c %d >> /var/log/intruder_alert
```

- To elevate the logging level:

```
sshd : ALL : severity emerg
```

- To deny access from /etc/hosts.allow:

```
sshd : .example.com : deny
```

Security Control in Oracle Cloud Infrastructure

- Private subnets – Disallow public IP addresses for instances in the subnet
- Security lists – Virtual firewalls that control packet level traffic for instances
 - Configured for subnets - all instances in the subnet have the same rules
 - Specify ingress and egress rules for traffic types
- Firewall rules – Control packet level traffic for instances
 - Configured on instances directly
 - Use `firewalld` in Oracle Linux 9
- Gateways and route tables – Control traffic between a Virtual Cloud Network (VCN) and destinations such as the Internet, on-premises network, Oracle Cloud Infrastructure services, and other VCNs
- IAM policies – Control access to the Oracle Cloud Infrastructure API and Console

Quiz



Which of the following has a specific purpose of allowing or denying access to network services?

- a. chroot jail
- b. firewall
- c. iptables
- d. TCP wrappers



Quiz



Which of the following has a specific purpose of allowing or denying access to network services?

- a. chroot jail
- b. firewall
- c. iptables
- d. TCP wrappers



Quiz



Which of the following statements are true? [Choose two]

- a. The default firewall service in Oracle Linux 9 is `firewalld`.
- b. When using `firewalld`, configuration changes do not require restart of `firewalld` service.
- c. With `iptables`, networks can be separated into different zones based on the level of trust.
- d. Both the `firewalld` and the `iptables` services include a GUI to make configuration changes.



Quiz



Which of the following statements are true?

- a. The default firewall service in Oracle Linux 9 is `firewalld`.
- b. When using `firewalld`, configuration changes do not require restart of `firewalld` service.
- c. With `iptables`, networks can be separated into different zones based on the level of trust.
- d. Both the `firewalld` and the `iptables` services include a GUI to make configuration changes.



Summary

In this lesson, you should have learned how to:

- Describe a `chroot` jail
- Use the `chroot` utility
- Describe packet-filtering firewalls
- Describe `firewalld`
- Configure `firewalld` packet filters
- Describe `iptables`
- Configure `iptables` packet filters
- Describe TCP wrappers
- Configure TCP wrappers
- Describe security control in Oracle Cloud Infrastructure



Practices for Lesson 16: Overview

This practice covers the following topics:

- Configuring a `chroot` jail
- Configuring a `chroot` jail for `ftp` users
- Exploring `firewalld`
- Configuring `firewalld`
- Configuring `iptables`
- Configuring a TCP wrapper
- Restoring VM configurations